

Topic 2: Proof Strategies ¹

ELEMENTS IN A MATHEMATICAL PROOF

A proof is the process of establishing the validity of a statement in a way that is consistent with the rules of logic. To prove a statement means to show that its conclusion follows from a set of premises, using a sequence of logical steps.

The initial statements or premises we *assume* to be true are called **assumptions**, typically introduced with phrases like “*Suppose...*” or “*Assume...*”. Any statement implied from these assumptions is only guaranteed to be true if the assumptions themselves are true.

The **conclusion** is the goal of the proof and the statement we aim to establish. Therefore, proving a statement involves demonstrating that its conclusion logically follows from its assumptions, given the truth of these assumptions. We may also rely on **definitions**, which are sets of logically equivalent statements that can be used interchangeably.

We will cover several proof methods: direct proof, proof by contrapositive, proof by contradiction, and mathematical induction. We will also review equivalence proofs and proofs with quantifiers.

DIRECT PROOF

Direct Proof is a way of proving statements that directly connects premises with implications. Let us consider the following example.

Example 1

We define an integer $x \in \mathbb{Z}$ is **even** if there exists an integer k such that $x = 2k$, and an integer $x \in \mathbb{Z}$ is **odd** if there exists an integer k such that $x = 2k + 1$. Show that if n is an odd integer, then $3n + 7$ is an even integer.

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Proof. When we analyze the proof, we need to find the premises and conclusions. The starting point is that “ n is an odd integer”, and the goal is “ $3n + 7$ is an even integer”, which can be written as

$$n \text{ is an odd integer} \Rightarrow 3n + 7 \text{ is an even integer.}$$

We also have definitions of odd and even integers that we can replace original statements with math expressions:

$$\exists k \in \mathbb{Z} \text{ such that } n = 2k + 1 \Rightarrow \exists h \in \mathbb{Z} \text{ such that } 3n + 7 = 2h.$$

Note that we use different letters k and h as the place holders as we do not know whether the two integers are the same.

Finally, we establish the connection between statements.

$$\exists k \in \mathbb{Z} \text{ such that } n = 2k + 1$$

$$\Rightarrow \exists k \in \mathbb{Z} \text{ such that } 3n = 3(2k + 1) = 2(3k) + 3$$

$$\Rightarrow \exists k \in \mathbb{Z} \text{ such that } 3n + 7 = 2(3k) + 3 + 7 = 2(3k) + 2 \times 5 = 2(3k + 5)$$

Let $h = 3k + 5$. Note that $h \in \mathbb{Z}$, and $3n + 7 = 2h$. Therefore, by definition, we can conclude that $3n + 7$ is an even integer. ■

PROOF BY CONTRAPOSITIVE

Proof by contrapositive exploits the fact that $P \Rightarrow Q$ and its contrapositive, $\neg Q \Rightarrow \neg P$, are logically equivalent. Instead of starting with P and then implying Q , you can start with $\neg Q$ and derive $\neg P$. Once you can establish $\neg Q \Rightarrow \neg P$, by logical equivalence, you have $P \Rightarrow Q$.

Let us prove the statement in Example 1 again using the proof by contrapositive approach.

Proof. Firstly, write down the statement to be proved.

$$3n + 7 \text{ is not an even integer} \Rightarrow n \text{ is not an odd integer.}$$

Rewrite the statement as the math expression.

$$\neg(\exists h \text{ such that } 3n + 7 = 2h) \Rightarrow \neg(\exists k \text{ such that } n = 2k + 1).$$

Be very careful here that we have NOT shown the property that if an integer is not an even number, then it is an odd number, and vice versa. Although it sounds very right (and it is indeed right), we can still not use this in our argument until it is proven true. We are only given the definition of an odd integer and an even integer.

We can have a detour to show the property that allows us to exchange oddity and evenness, but let's keep working with the contrapositives. For the previous statement, equivalently,

$$\forall h \in \mathbb{Z}, 3n + 7 \neq 2h \Rightarrow \forall k \in \mathbb{Z}, n \neq 2k + 1.$$

Then we establish the connection between statements.

$$\begin{aligned} &\forall h \in \mathbb{Z}, 3n + 7 \neq 2h \\ \Rightarrow &\forall h \in \mathbb{Z}, 3n \neq 2h - 7 \\ \Rightarrow &\forall h \in \mathbb{Z}, n \neq \frac{2h - 7}{3} \end{aligned}$$

As the statement is true for every $h \in \mathbb{Z}$, it is true for every $k \in \mathbb{Z}$ that $h = 3k + 5$. Hence,

$$\forall k \in \mathbb{Z}, n \neq \frac{2(3k + 5) - 7}{3} = \frac{6k + 3}{3} = 2k + 1.$$

Therefore, we showed the conclusion of the contrapositive, which implies that the original statement is correct. ■

It may seem very unwise to prove by contrapositive in this case. However, depending on the situation, proving oppositely can be easier than a direct proof. For example, prove by contrapositive will definitely be easier for this statement:

$$3n + 7 \text{ is not an even integer} \Rightarrow n \text{ is not an odd integer.}$$

Most of the time, it requires practice to see which is the easier way to prove.

PROOF BY CONTRADICTION

Proof by contradiction (or *reductio ad absurdum*) is also a useful proving technique, especially for proofs that do not have clear implication forms as $P(x) \Rightarrow Q(x)$.

We start by **assuming** that the statement R we are proving is *false*. (We usually begin with the phrase “*Assume, to the contrary, R is false.*”) Then we show that this assumption leads to some statements that contradict what we know to be true, resulting in a logical contradiction. Proof by contradiction is essentially showing $\neg R \Rightarrow (P \wedge \neg P)$. As $(P \wedge \neg P)$ is a contradiction, the whole statement can only be true if $\neg R$ is false, which implies that the statement we want to show R is true.

Proof by contradiction is useful to show the negative-sounding statements. See the following example.

Example 2

Prove that there is no smallest positive real number.

Proof. Assume, to the contrary, that there exists a smallest positive real number. Call this real number r . Then, we try to derive some statements that contradict to something that we know to be true.

Let $s = \frac{r}{2}$. Note that $s \in \mathbb{R}$ and $0 < s < r$. Since we indeed find a positive real number smaller than r , r is not the smallest positive real number, which leads to a contradiction. ■

In the prove above, we assume the negation of the statement we are proving is true. We can easily see that

$\neg R$: there is some r that is the smallest positive real number.

It immediately implies

$S : s = \frac{r}{2}$ is not smaller than r because $s \neq r$.

But we also show that

$\neg S : s = \frac{r}{2}$ is smaller than r .

Therefore, we have shown that

$$\neg R \Rightarrow (\neg S \wedge S).$$

where $\neg S \wedge S$ forms a contradiction, and it is reached because we assume $\neg R$ is true.

Therefore, $\neg R$ must be false. In other words, R is true.

EQUIVALENCE PROOF (PROVING $P \Leftrightarrow Q$)

$P \Leftrightarrow Q$ and $(P \Rightarrow Q) \wedge (Q \Rightarrow P)$ are logically equivalent. Therefore, to show that any two statements are logically equivalent, you can just **show the two statements imply each other**.

Example 3

Suppose A, B are non-empty sets. Show that $A = B$ if and only if the elements in A and B are identical.

For this statement, you have two directions to prove: **sufficiency** ($\text{identical} \Rightarrow A = B$) and **necessity** ($A = B \Rightarrow \text{identical}$). Be very careful that when you are proving one direction, do not take the assumptions or premises you use for the other direction.

Proof. We firstly show sufficiency using a direct proof. Since the elements in the two sets are identical, it is immediate that for every element in A , it must in B , and *vice versa*. By the definition of subsets, this implies $A \subset B$ and $B \subset A$. Consequently, by the definition of equal sets, we have $A = B$.

We now show necessity using a proof by contradiction. Suppose $A = B$, but A and B do not have identical elements.² Then, without loss of generality, there exists $x \in A$ such that $x \notin B$. By the definition of subsets, this implies $A \not\subset B$. But by the definition of set equality, $A = B$ if and only if $A \subset B$ and $B \subset A$. Since $A \not\subset B$, it follows that $A \neq B$, contradicting our assumption. ■

In some cases, we may want to prove a sequence of equivalent statements, like $P \Leftrightarrow Q \Leftrightarrow R$. One strategy is to prove pairwise equivalences (e.g., $P \Leftrightarrow Q$ and $Q \Leftrightarrow R$) and then use the transitivity of equivalence to conclude the full chain. Alternatively, we can prove the statement by forming a full implication circle:

$$(P \Rightarrow Q) \wedge (Q \Rightarrow R) \wedge (R \Rightarrow P).$$

This ensures that each statement leads to the next, and the cycle eventually returns to its starting point.

²Recall that $P \Rightarrow Q$ and $\neg(P \wedge \neg Q)$ are logically equivalent.

Why does this work? Because implication is transitive. For example,

$$((P \Rightarrow Q) \wedge (Q \Rightarrow R)) \Rightarrow (P \Rightarrow R).$$

and together with $R \Rightarrow P$, we obtain $P \Leftrightarrow R$.

Example 4

Suppose A and B are sets. Show the following statements are equivalent.

1. $A \subset B$.
2. $A \cup B = B$.
3. $A \cap B = A$.

PROOF WITH QUANTIFIERS

There are three logical quantifiers: universal quantifier (“for all”), existential quantifier (“there exists”), and uniqueness existential quantifier (“there exists unique”). Here are the proof strategies for each.

(1) **“For all” statements** $\forall x, P(x)$

To show a “for all” statement, you have two approaches.

- (a) Direct proof: Pick any x , and show that $P(x)$ is true. Then claim that “since x is arbitrarily picked, the statement is true for any x .” Be *very careful* that you are really arbitrarily picking x without any implicit conditions.
- (b) Proof by contradiction: Assume that there exists x such that $P(x)$ is false. Then show that this leads to a contradiction.

(2) **“There exists” statements** $\exists x$ s.t. $P(x)$.

There is nothing fancy—just try to find some x that satisfies $P(x)$.

(3) **“There exists unique” statements** $\exists! x$ s.t. $P(x)$.

Now you not only need to check that there exists some x makes $P(x)$ true, but you also need to make sure that this x you find is the one and only one x qualified. Here are the steps for uniqueness proofs.

- (i) Find one x , and show that $P(x)$ is true.
- (ii) Assume there exists another $y \neq x$ such that $P(y)$ is true.
- (iii) Show that this leads to a contradiction.

Example 5

Show that there exists a unique x such that $(x - 3)^2 = 0$.

Proof. Firstly, it is easy to see that $x = 3$ satisfies the requirement. Then assume that there exists $y \neq 3$ such that $(y - 3)^2 = 0$. As $y \neq 3$, either $y > 3$ or $y < 3$.

Consider the first case: $y > 3$. Then $y - 3 > 0$, so $(y - 3)^2 > 0$, which contradicts $(y - 3)^2 = 0$. Next, consider the second case: $y < 3$. Then $y - 3 < 0$, so $(y - 3)^2 > 0$, which contradicts $(y - 3)^2 = 0$. Both cases of $y > 3$ or $y < 3$ will lead to contradiction. So, when $y \neq 3$, $(y - 3)^2 = 0$ cannot be true. It proves the uniqueness. ■

Next, we come back to the proposition we mentioned earlier.

Proposition 1

$\emptyset$ is a subset of any set S .

This proposition looks very strange to prove directly. Recall the definition of $A \subset S$ is “ $\forall x \in A, x \in A \Rightarrow x \in S$.” Nonetheless, there is nothing in an empty set! Although it is possible to show the statement directly (as we will show below), it may seem more intuitive to prove by contradiction.

Proof. We prove the proposition using two methods.

(1) Direct proof

Note that $\emptyset \subset S$ if and only if $\forall x \in \emptyset, x \in \emptyset \Rightarrow x \in S$. Since $x \in \emptyset$ is always false, the whole statement is always true by the truth table of the implication operator ($\Rightarrow$). This is true for any chosen S . Therefore $\emptyset \subset S$ for any set S .

(2) Proof by contradiction

Assume the opposite: $\exists x \in \emptyset$ such that $x \in \emptyset$ and $x \notin S$. However, as there can be no element in the empty set. Therefore, $x \in \emptyset$ and $x \notin S$ cannot be true, and we arrive at a contradiction. Therefore, the original statement is true. ■

MATHEMATICAL INDUCTION

Consider the following statement from high school mathematics:

$$\forall n \in \mathbb{N}, 1 + 2 + \cdots + n \equiv \sum_{k=1}^n k = \frac{n(n+1)}{2}.$$

How do you check this statement is true?

Recall that for this “for all” statement, we can use a direct proof or a proof by contradiction. Notice that the direct proof alone does not work because we need to check *all* n and make sure this statement is true, but there are infinitely many n , so we cannot check all of them.

For this type of proof, going over any natural numbers, we can use the **mathematical induction**.³

Theorem 1: The Principle of Mathematical Induction

For each positive integer n , let $P(n)$ be a statement.

If

(1) $P(1)$ is true, and

(2) For every positive integer k , $P(k)$ implies $P(k+1)$,

then $P(n)$ is true for every positive integer n .

Proof. We can prove it by contradiction.

Assume the principle of mathematical induction is not true. In other words, assume (1) $P(1)$ is true and (2) For every positive integer k , $P(k)$ implies $P(k+1)$ are both true, but there exist some $x \in \mathbb{N}$ such that $P(x)$ is false.

Let x_s be the smallest integer such that $P(x_s)$ is false. When $x_s = 1$, it contradicts to the assumption that $P(1)$ is true. When $x_s > 1$, the statement $P(x_s - 1) \Rightarrow P(x_s)$ will be false. Since the assumption leads to contradictions, there must be no $x \in \mathbb{N}$ such that $P(x)$ is false. ■

³In fact, the principle of mathematical induction requires a set of axioms on natural numbers. Refer to the Peano axioms if you are interested.

Example 6

Show that for every $n \in \mathbb{N}$,

$$1 + 2 + \cdots + n \equiv \sum_{k=1}^n k = \frac{n(n+1)}{2}.$$

Proof. To apply mathematical induction, we need to check two conditions:

(1) $P(1)$ is true. It easy to see that $1 = \frac{1(1+1)}{2}$.

(2) For every positive integer k , $P(k)$ implies $P(k+1)$. We need to check the following statement: for any $k \in \mathbb{N}$, if $\sum_{k'=1}^k k' = \frac{k(k+1)}{2}$, then $\sum_{k'=1}^{k+1} k' = \frac{(k+1)(k+2)}{2}$.

Note that $\sum_{k'=1}^k k' = \frac{k(k+1)}{2}$. Hence

$$\begin{aligned} & \sum_{k'=1}^{k+1} k' \\ &= \sum_{k'=1}^k k' + (k+1) && \text{(by definition of summation)} \\ &= \frac{k(k+1)}{2} + (k+1) && \text{(by the premise in (2))} \\ &= \frac{k(k+1)}{2} + \frac{2(k+1)}{2} && \text{(by arithmetic)} \\ &= \frac{(k+1)(k+2)}{2}. \end{aligned}$$

Since conditions (1) and (2) are true, by mathematical induction, the statement is true. ■

Mathematical induction is very useful when dealing with statement about all natural numbers. However, mathematical induction can *only* be used to prove whether a statement holds for all natural numbers. If you want to apply mathematical induction, make sure you can write the statement as stated in the Principle of Mathematical Induction.

REFERENCES

Smith, D., Eggen, M., and Andre, R. (2010). *A Transition to Advanced Mathematics*. Cengage Learning.